

PROJECT SYNOPSIS

Crop Disease Detection Using Convolutional Neural Networks (CNN)

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1. Title of the Project

Crop Disease Detection Using Convolutional Neural Networks (CNN) with Hierarchical Binary Gate Verification.

2. Objective & Problem Statement

Agricultural crop diseases cause up to 40% annual global crop yield loss. Traditional manual scouting by agricultural experts is slow, expensive, and unavailable to remote farmers. The objective of this project is to construct an end-to-end automated computer vision system that instantly identifies crop species, classifies 38 plant diseases/healthy states, and provides agronomic treatment advisory.

3. System Architecture & Methodology

The system implements a **Two-Stage Hierarchical Deep Learning Pipeline**:

- **Stage 1 (Approach 1: Binary Leaf Gate Model)**: Filters non-leaf or out-of-focus background uploads using a binary CNN (Is Leaf vs Non-Leaf).
- **Stage 2 (38-Class Multiclass CNN)**: Evaluates valid plant leaves using a 4-block Conv2D network (Conv2D -> BatchNorm -> MaxPool -> Dropout -> Softmax).
- **Stage 3 (Web Portal & Advisory Engine)**: Delivers predictions via a responsive Flask web interface with treatment recommendations.

4. Key Technical Specifications

Component	Specification
Dataset	PlantVillage Dataset (54,306 RGB images, 38 classes, 14 crop species)
Programming Language	Python 3.12
Deep Learning Framework	TensorFlow 2.18 / Keras
Stage 1 Classifier	Approach 1 Binary Gate CNN (Sigmoid Activation)
Stage 2 Classifier	Baseline 4-Block Conv2D CNN (Softmax Activation)
Web Application	Flask 3.0 Backend with HTML5/CSS3/JavaScript Frontend

5. Project Deliverables

1. Modular Python ML Source Code (src/)
2. Trained Model Weights (model/binary_gate_model.h5 & model/crop_disease_cnn_best.keras)
3. Flask Web Application (app.py)
4. Academic Viva & Examination Guide (viva_prep.md)
5. Comprehensive PDF Project Report & Official Synopsis Document